Forces shaping genetic diversity

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An introduction to coalescent theory

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Genetic diversity

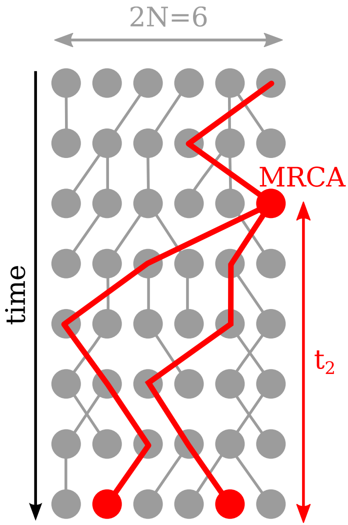
We have seen that:

- Over time, genetic drift is reducing genetic diversity.
- The reason we still observe genetic diversity are new mutations.

Question: How much genetic diversity does a population have?

This is best understood using coalescent theory.

Intro to Coalescent Theory



Coalescent theory

A population genetic theory that considers the history of a sample **backward in time**.

Coalescent event

If two sampled lineages have the same parent in the previous generation.

Probability to coalesce

Under random mating, two lineages coalesce in the previous generation with probability

$$\mathbb{P}(2 \text{ individuals coalesce}) = \frac{1}{2N}$$

Expected time t_2 until two lineages coalesce (time to the Most Recent Common Ancestor, MRCA):

$$\mathbb{E}[t_2] = 2N \text{ generations.}$$

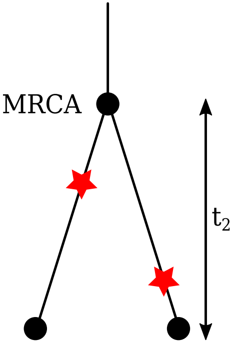
Mutations on the Coalescent

Expected number of mutations between two samples

If the mutation rate per generation per lineage is μ, the expected number of mutational differences d_{ij} between two samples i and j is thus

$$\mathbb{E}[d_{ij}] = 2 \cdot \mathbb{E}[t_2]\mu = 4N\mu$$

Note: the population size N and the mutation rate μ have the same effect on genetic diversity!



Definition: θ

$\theta := 4N\mu$ is an important population genetics parameter characterizing the expected genetic diversity of a population.

Estimating θ

Infinite Sites Model (ISM)

Assumes that each mutation hits a different base pair. This is realistic if $\theta \ll 1$.

The Tajima estimator of θ under ISM

Since we expect $\theta = 4N\mu$ mutations between two samples, the average number of differences π between individuals is an estimate of θ.

$$\hat{\theta}_T = \pi = \frac{1}{n(n-1)/2} \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} d_{ij},$$

where $n(n-1)/2$ is the number of pairs among n samples.

Note: While $\mathbb{E}[\pi] = \theta$, any particular value of π will unlikely be exactely θ!

This is why we call π and *estimator* of θ and write it as $\hat{\theta}$ or $\hat{\theta}_T$ to indicate that it is the *Tajima* estimator.

Estimating θ

Example: Sequence data from four samples (lineages)

```
Sample 1:  aggtatgcta gaaccctaga aagacacaga gatagacaag tgttaattca
Sample 2:  aggcatgcta gaaccctaga aagacgcaga gatagacaag tgttcattca
Sample 3:  aggcatgcta gaaccctaga aagacacaga tatagacaag tgttaattca
Sample 4:  aggtatgctg gaaccctaga aagacacaga tatagacaag tgttaattca
```

$$\hat{\theta}_T = \pi = \frac{1}{n(n-1)/2} \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} d_{ij} = \frac{d_{12} + d_{13} + d_{14} + d_{23} + d_{24} + d_{34}}{6}$$

$$= \frac{3+2+2+3+5+2}{6} = \frac{17}{6} = 2.83$$

Note: to compare numbers between studies, it is best to report π / site:

$$\hat{\theta}_T = \pi = \frac{3+2+2+3+5+2}{6} \cdot \frac{1}{50} = \frac{17}{6 \cdot 50} = \frac{17}{300} = 0.057$$

Effective population sizes

We have seen that $\mathbb{E}[d_{ij}] = 4N\mu$. For humans: $\mathbb{E}[d_{ij}] = 4 \cdot 8 \cdot 10^9 \cdot 10^{-8} = 320$.
Bu wait: human heterozygosity is $\approx 0.1\%$, not 100%!

All these theoretical results are for an idealized, simple model of a constant population. Yet, we can force it to work for many non-standard models.

The concept of effective population size (N_e)

Effective population size: An adjusted population size we can use under a WF-model to accurately predict properties of non-WF populations.

Example (Varying population size)

If population sizes vary through time as: $N_0, N_1, N_2, \ldots, N_t$, a WF-model with

$$N_e = 1/\left[\frac{1}{t}\left(\frac{1}{N_0} + \frac{1}{N_1} + \ldots + \frac{1}{N_{t-1}}\right)\right]$$

still predicts, for example, the decrease in heterozygosity through time perfectly.

Coalescence with multiple samples

Probability of coalescent

$$P(\text{at least one coalescent event}) = \binom{k}{2}\frac{1}{2N} = \frac{k(k-1)}{4N}$$

Intuitive explanation

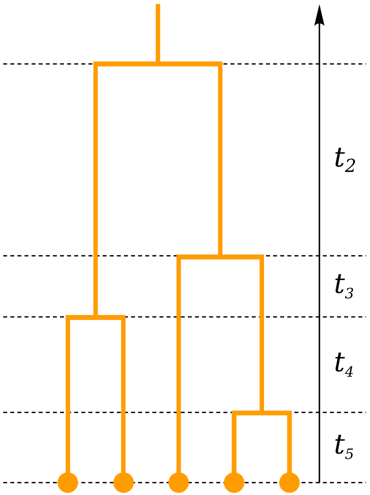
Probability of coalescence among k lineages = probability of coalescence among two lineages $\frac{1}{2N}$ times the number of possible pairs $\binom{k}{2}$.

Expected time t_k until first coalescent event among k lineages

The expected waiting time until an event occurs the first time is given by the inverse of the probability of the event!

$$\mathbb{E}[t_k] = \frac{1}{\binom{k}{2}\frac{1}{2N}} = \frac{2N}{\binom{k}{2}} = \frac{4N}{k(k-1)}$$

Expected genealogy of n samples (lineages)



$$\mathbb{E}[t_k] = \frac{1}{\binom{k}{2}\frac{1}{2N}} = \frac{2N}{\binom{k}{2}} = \frac{4N}{k(k-1)}$$

$$\mathbb{E}[t_2] = \frac{2N}{\binom{2}{2}} = 2N$$

$$\mathbb{E}[t_3] = \frac{2N}{\binom{3}{2}} = \frac{2N}{3}$$

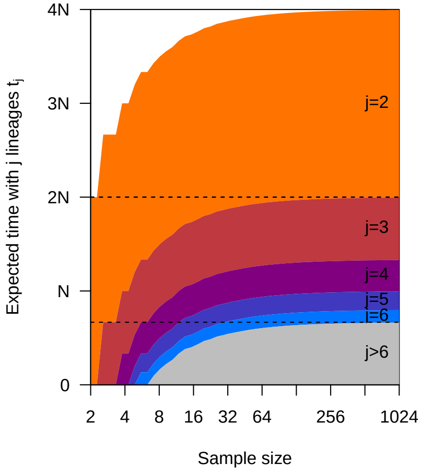
$$\mathbb{E}[t_4] = \frac{2N}{\binom{4}{2}} = \frac{2N}{6}$$

$$\mathbb{E}[t_5] = \frac{2N}{\binom{5}{2}} = \frac{2N}{10}$$

Note: Tree is dominated by epochs with few lineages!

Note: Relative proportions are independent of N!

Expected genealogy of n samples (lineages)



$$\mathbb{E}[t_k] = \frac{1}{\binom{k}{2}\frac{1}{2N}} = \frac{2N}{\binom{k}{2}} = \frac{4N}{k(k-1)}$$

Expected total tree height

$$\mathbb{E}[t_{MRCA}] = \sum_{k=2}^{n} \mathbb{E}[t_k]$$

$$= \sum_{k=2}^{n} \frac{4N}{k(k-1)}$$

$$= 4N\left(1 - \frac{1}{n}\right)$$

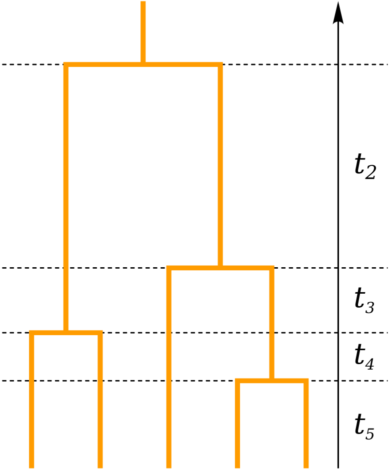
Note: Tree is dominated by epochs with few lineages!

Note: Relative proportions are independent of N!

Expected genealogy of n samples (lineages)

Expected total length L_n of the genealogy

The expected total length of a genealogy of n samples, L_n is the sum of the expected epoch length times the number of lineages during that epoch.



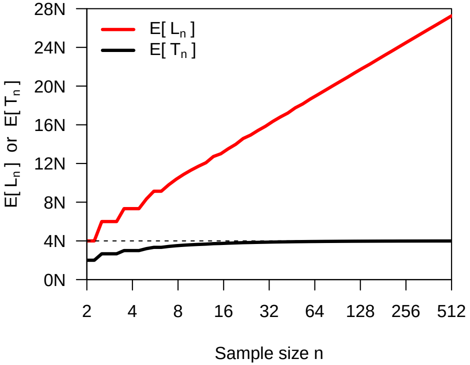
$$\begin{aligned}\mathbb{E}[L_n] &= \sum_{k=2}^{n} \mathbb{E}[t_k]k \\ &= \sum_{k=2}^{n} \frac{4N}{k(k-1)}k \\ &= 4N\sum_{k=2}^{n} \frac{1}{k-1} \\ &= 4N\sum_{k=1}^{n-1} \frac{1}{k}\end{aligned}$$

Expected genealogy of n samples (lineages)

Height versus length of a genealogy of n samples

$$\mathbb{E}[T_n] = 4N\left(1 - \frac{1}{n}\right)$$

$$\mathbb{E}[L_n] = 4N\sum_{k=1}^{n-1}\frac{1}{k}$$



Note: Adding additional samples does increase the expected tree height only marginally, but increases the tree length a lot.

Mutations on the Coalescent

Expected number of mutations

The expected number of mutations $\mathbb{E}[S]$ on a genealogy is given by the product of the expected length of the genealogy $\mathbb{E}[L_n]$ and the mutation rate μ:

$$\mathbb{E}[S] = \mathbb{E}[L_n]\mu = 4N\mu \sum_{k=1}^{n-1} \frac{1}{k} = \theta \sum_{k=1}^{n-1} \frac{1}{k}$$

Infinite Sites Model (ISM)

Assumes that each mutation hits a different base pair. This is realistic if $\theta \ll 1$.

The Watterson estimator of θ under ISM

Under ISM, the number of mutations = to the number of polymorphic (called segregating) sites S among samples. S can thus be used to estimate θ:

$$\mathbb{E}[S] = \theta \sum_{k=1}^{n-1} \frac{1}{k} \qquad \Rightarrow \qquad \hat{\theta}_W = \frac{S}{\sum_{k=1}^{n-1} \frac{1}{k}}$$

Example: the Watterson estimator of θ

Example: Sequence data from four samples (lineages)

```
Sample 1:  aggtatgcta gaaccctaga aagacacaga gatagacaag tgttaattca
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Sample 3:  aggcatgcta gaaccctaga aagacacaga tatagacaag tgttaattca
Sample 4:  aggtatgctg gaaccctaga aagacacaga tatagacaag tgttaattca
```

$$\hat{\theta}_W = \frac{S}{\sum_{k=1}^{n-1} \frac{1}{k}} = \frac{5}{1 + \frac{1}{2} + \frac{1}{3}} = \frac{5}{\frac{11}{6}} = \frac{30}{11} = 2.72$$

Note: to compare numbers between studies, it is best to report $\hat{\theta}_W$ / site:

$$\hat{\theta}_W = \frac{30}{11 * 50} = \frac{30}{550} = 0.055$$

Comparison with $\hat{\theta}_T$

$\hat{\theta}_W = 0.055$ is close yet not identical to $\hat{\theta}_T = 0.057$ as both are *estimates* of θ!

The Site Frequency Spectrum (SFS)

The SFS f tabulates the sample allele frequencies of all mutations.

Each entry f_1, f_2, ..., f_{n-1} is the number (or proportion) of mutations at which 1, 2, ..., $n-1$ **derived** alleles have been observed in a sample of size n.

Example: Sequence data from four samples (lineages)

```
Sample 1:  aggtatgcta gaaccctaga aagacacaga gatagacaag tgttaattca
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Sample 4:  aggtatgctg gaaccctaga aagacacaga tatagacaag tgttaattca
```
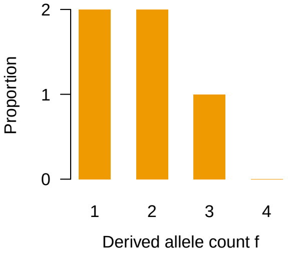

There are 5 mutations in total. Of those,

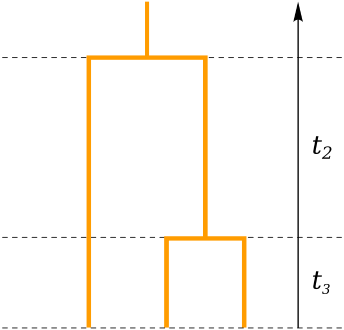
$f_1 = 2$ have 1 derived allele copy,

$f_2 = 2$ have 2 derived allele copies,

$f_3 = 1$ have 3 derived allele copies.



The Site Frequency Spectrum (SFS)



Note: We expect the same number of singletons among 2 and 3 samples!

Expected SFS when $n = 2$

All mutations are singletons:

$$\mathbb{E}[f_1] = 2\,\mathbb{E}[t_2]\mu = 4N\mu = \theta$$

Sample of $n = 3$

Mutations falling on a branch with only one leave result in singletons.

$$\begin{aligned}\mathbb{E}[f_1] &= (3\,\mathbb{E}[t_3] + \mathbb{E}[t_2])\mu \\ &= \left(3\frac{2N}{3} + 2N\right)\mu = 4N\mu = \theta\end{aligned}$$

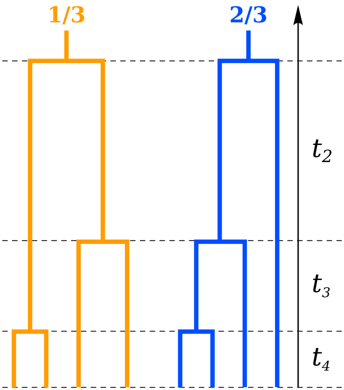
Mutations falling on branches with two leaves result in doubletons.

$$\mathbb{E}[f_2] = \mathbb{E}[t_2]\mu = 2N\mu = \frac{\theta}{2}$$

The Site Frequency Spectrum (SFS)

Expected SFS when $n = 4$

There are two different topologies with probabilities $\frac{1}{3}$ and $\frac{2}{3}$, respectively!



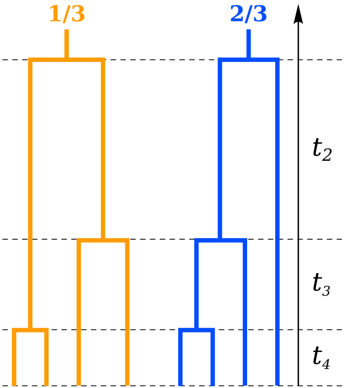
$$
\begin{aligned}
\mathbb{E}[f_1] &= \frac{1}{3}(4\,\mathbb{E}[t_4] + 2\,\mathbb{E}[t_3])\mu \\
&+ \frac{2}{3}(4\,\mathbb{E}[t_4] + 2\,\mathbb{E}[t_3] + \mathbb{E}[t_2])\mu \\
&= \left(4\,\mathbb{E}[t_4] + 2\,\mathbb{E}[t_3] + \frac{2}{3}\,\mathbb{E}[t_2]\right)\mu \\
&= \left(4\frac{2N}{6} + 2\frac{2N}{3} + \frac{2}{3}2N\right)\mu \\
&= \left(\frac{8}{6} + \frac{8}{6} + \frac{8}{6}\right)N\mu = 4N\mu = \theta
\end{aligned}
$$

Note: We expect the same number of singletons among 2, 3 and 4 samples!

The Site Frequency Spectrum (SFS)

Expected SFS when $n = 4$

There are two different topologies with probabilities $\frac{1}{3}$ and $\frac{2}{3}$, respectively!



$$\begin{aligned}\mathbb{E}[f_2] &= \left(\frac{1}{3}(\mathbb{E}[t_3] + 2\,\mathbb{E}[t_2]) + \frac{2}{3}\,\mathbb{E}[t_3]\right)\mu \\ &= \left(\mathbb{E}[t_3] + \frac{2}{3}\,\mathbb{E}[t_2]\right)\mu \\ &= \left(\frac{2N}{3} + \frac{2}{3}2N\right)\mu = 2N\mu = \frac{\theta}{2}\end{aligned}$$

Note: We expect the same number of doubletons among 3 and 4 samples!

$$\mathbb{E}[f_3] = \frac{2}{3}\,\mathbb{E}[t_2]\mu = \frac{2}{3}2N\mu = \frac{4N\mu}{3} = \frac{\theta}{3}$$

The Site Frequency Spectrum (SFS)

Expected SFS for n samples (lineages)

So far we found:

- $\mathbb{E}[f_1] = \theta$ among 2, 3 or 4 samples.
- $\mathbb{E}[f_2] = \frac{\theta}{2}$ among 3 or 4 samples.
- $\mathbb{E}[f_3] = \frac{\theta}{3}$ among 4 samples.

General rule

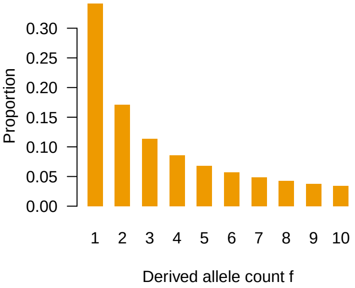
It is possible to prove that

$$\mathbb{E}[f_k] = \frac{\theta}{k}$$

Relative proportions

To get the expected relative proportions, we just have to divide by the total number of mutations:

$$\frac{\mathbb{E}[f_k]}{\mathbb{E}[S]} = \frac{\frac{\theta}{k}}{\theta \sum_{j=1}^{n-1} \frac{1}{j}} = \frac{\frac{1}{k}}{\sum_{j=1}^{n-1} \frac{1}{j}} = \frac{1}{k \sum_{j=1}^{n-1} \frac{1}{j}}$$



Wahlund Effect and F_{ST}

Wahlund Effect

The decrease in heterozygosity due to population subdivision.

Population subdivision (or Wahlund effect) is quantified by $F_{ST} = \dfrac{H_T - H_S}{H_T}$.

Expected Hetertozygosity within Subpopulations H_S

Under HWE and if the subpopulations are of equal size:

$$H_S = \frac{H_1 + H_2}{2} = f_1(1 - f_1) + f_2(1 - f_2).$$

$\Rightarrow$ the heterozygosity expected among samples from the subdivided population.

Expected Total Hetertozygosity H_T

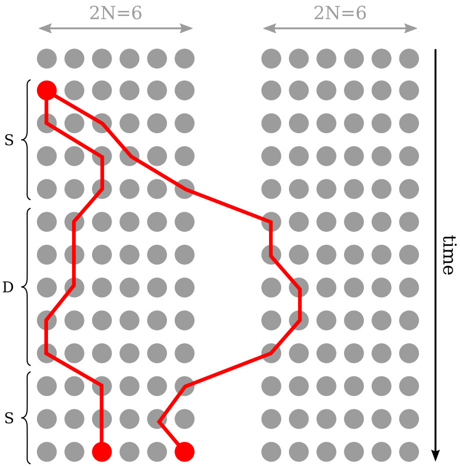
Expected heterozygosity if the total population was in HWE: $H_T = 2f_T(1 - f_T)$

If the subpopulations are of equal size, $f_T = \dfrac{f_1 + f_2}{2}$.

$\Rightarrow$ $F_{ST} = 0$ when subpopulations do not differ in allele frequency.

$\Rightarrow$ $F_{ST} = 1$ when subpopulations are fixed for different alleles.

Coalescent process with migration



Probability of a migration event

m: probability that a lineage will migrate per generation.

Migration or Coalescence?

Consider two lineages currently in the same population.

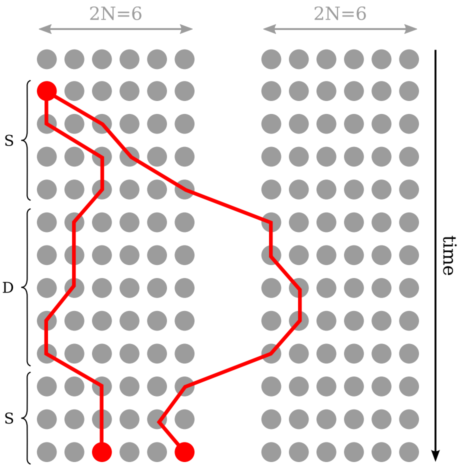
Probability of coalescence: $P_c = \frac{1}{2N}$.

Probability that one will migrate: $P_m = 2m(1-m) \approx 2m$.

Probability that they coalesce before either of them migrates:

$$P_{c<m} = \frac{P_c}{P_c + P_m}$$

Coalescent process with migration



Expected time to MRCA

If the two lineages are in different populations:

$$\begin{aligned}\mathbb{E}[t_D] &= \mathbb{E}[t_m] + \mathbb{E}[t_S] \\ &= \frac{1}{2m} + \mathbb{E}[t_S]\end{aligned}$$

If the two lineages are in the same population:

$$\begin{aligned}\mathbb{E}[t_S] &= \mathbb{E}[t_{m \text{ or } c}] + P_{m<c}\,\mathbb{E}[t_D] \\ &= \frac{1}{\frac{1}{2N} + 2m} + \frac{P_m}{P_c + P_m}\,\mathbb{E}[t_D] \\ &= \frac{1}{\frac{1}{2N} + 2m} + \frac{2m}{\frac{1}{2N} + 2m}\,\mathbb{E}[t_D] \\ &= \frac{1}{\frac{1}{2N} + 2m}\left(1 + 2m\,\mathbb{E}[t_D]\right)\end{aligned}$$

Coalescent process with migration

Expected time to MRCA

We thus have $\mathbb{E}[t_D] = \frac{1}{2m} + \mathbb{E}[t_S]$ and $\mathbb{E}[t_S] = \frac{1}{\frac{1}{2N}+2m}\left(1 + 2m\,\mathbb{E}[t_D]\right)$

We can solve this system of two equations easily:

$$\mathbb{E}[t_S] = \frac{1}{\frac{1}{2N}+2m}\left(1 + 2m\left(\frac{1}{2m} + \mathbb{E}[t_S]\right)\right)$$

$$= \frac{1}{\frac{1}{2N}+2m}\left(2 + 2m\,\mathbb{E}[t_s]\right)$$

$$\mathbb{E}[t_S]\left(1 - \frac{2m}{\frac{1}{2N}+2m}\right) = \frac{2}{\frac{1}{2N}+2m}$$

$$\mathbb{E}[t_S] = \frac{\frac{2}{\frac{1}{2N}+2m}}{1 - \frac{2m}{\frac{1}{2N}+2m}} = \frac{\frac{2}{\frac{1}{2N}+2m}}{\frac{\frac{1}{2N}+2m-2m}{\frac{1}{2N}+2m}} = \frac{2}{\frac{1}{2N}} = 4N$$

$$\mathbb{E}[t_D] = \frac{1}{2m} + 4N$$

Note: $\mathbb{E}[t_S]$ is independent of m, but twice as in the absence of migration!

Expected F_{ST} in the presence of migration

Expected Heterozygosity within subpopulation H_S

$\mathbb{E}[H_S] = 2\mu\,\mathbb{E}[t_S] = 2\mu \cdot 4N = 8N\mu = 2\theta$

Expected total Heterozygosity H_T

A random pair of lineages is in 50% of the cases from the same subpopulation, and in 50% from different subpopulations.

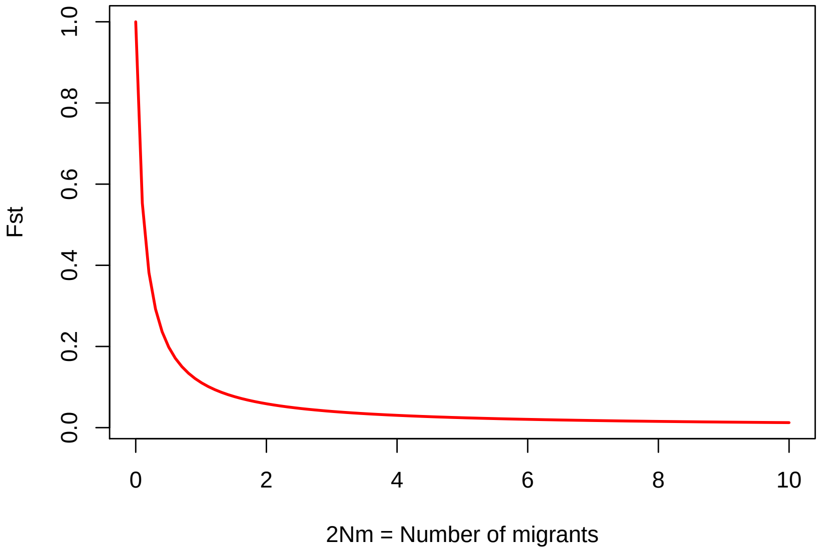
$$\mathbb{E}[H_T] = \mu\frac{2\,\mathbb{E}[t_S] + 2\,\mathbb{E}[t_D]}{2} = (\mathbb{E}[t_S]+\mathbb{E}[t_D])\mu = (4N+\frac{1}{2m}+4N)\mu = \frac{\mu}{2m}+8N\mu$$

Expected F_{ST}

$$\mathbb{E}[F_{ST}] = \frac{\mathbb{E}[H_T] - \mathbb{E}[H_S]}{\mathbb{E}[H_T]} = \frac{\frac{\mu}{2m} + 8N\mu - 8N\mu}{\frac{\mu}{2m} + 8N\mu} = \frac{\frac{\mu}{2m}}{\frac{\mu}{2m} + 8N\mu}$$

$$= \frac{\frac{1}{2m}}{\frac{1}{2m} + 8N} = \frac{1}{1 + 16Nm}$$

Coalescent process with migration



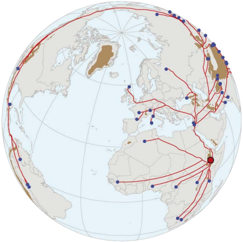
Note: already very few migrants inhibit strong divergence!

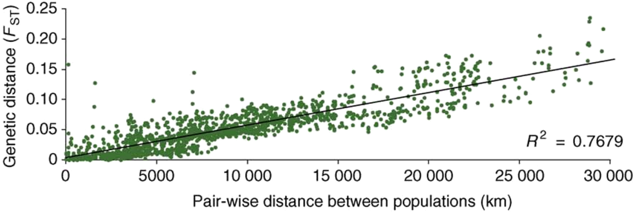
Isolation by distance

Isolation by distance

Since the effective migration often depends on the distance between populations, closer populations tend to show smaller F_{ST} than more distant populations.

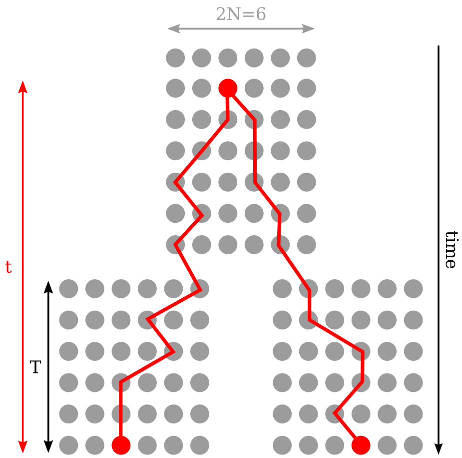
Example: F_{ST} between human populations





Handley *et al.* 2007

Coalescent with divergence



Expected time to MRCA

Assuming all populations to have equal sizes, $\mathbb{E}[t_S] = 2N$

Going backward in time, two lineages sampled from different subpopulations can only coalesce after the divergence event.

$\mathbb{E}[t_D] = T + 2N$

Expected F_{ST} under divergence

Expected Heterozygosity within subpopulation H_S

$\mathbb{E}[H_S] = 2\,\mathbb{E}[t_S]\mu = 2 \cdot 2N\mu = 4N\mu = \theta$

Expected total Heterozygosity H_T

A random pair of lineages is in 50% of the cases from the same subpopulation, and in 50% from different subpopulations.

$$\mathbb{E}[H_T] = \mu\frac{2\,\mathbb{E}[t_S] + 2\,\mathbb{E}[t_D]}{2} = (\mathbb{E}[t_S] + \mathbb{E}[t_D])\mu = (T + 2N + 2N)\mu = T\mu + 4N\mu$$

Expected F_{ST}

$$\mathbb{E}[F_{ST}] = \frac{\mathbb{E}[H_T] - \mathbb{E}[H_S]}{\mathbb{E}[H_T]} = \frac{T\mu + 4N\mu - 4N\mu}{T\mu + 4N\mu} = \frac{T\mu}{T\mu + 4N\mu} = \frac{T}{T + 4N}$$

Simulating Genealogies

Distribution of coalescent times

The probability that k lineages coalesce t generations ago is

$$\mathbb{P}(t_k = t) = \left(1 - \frac{k(k-1)}{4N}\right)^{t-1} \frac{k(k-1)}{4N}$$

Using the approximation

$$(1-x)^t \approx e^{-xt}$$

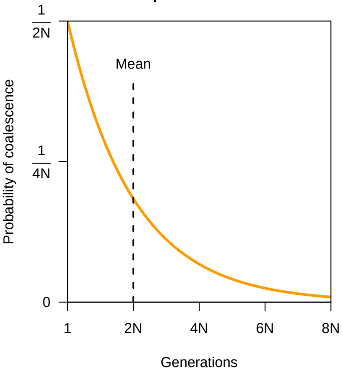
we can rewrite this as

$$\mathbb{P}(t_k = t) = \frac{k(k-1)}{4N} e^{-\frac{k(k-1)}{4N}(t-1)}$$

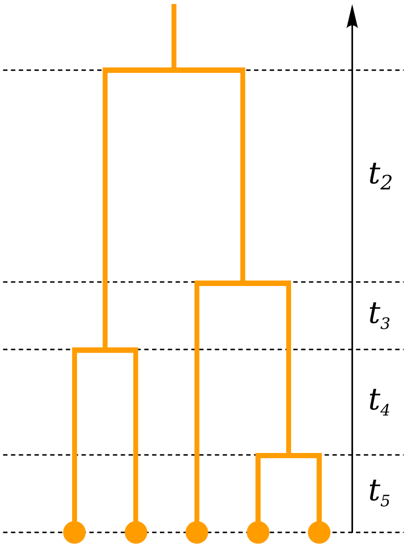
Thus t_k is exponentially distributed as

$$t_k \sim \mathrm{Exp}\left(\frac{k(k-1)}{4N}\right)$$

Example for t_2



Simulating a genealogy of n samples



$$t_k \sim \text{Exp}\left(\frac{k(k-1)}{4N}\right)$$

Simulate $t_2 \sim \text{Exp}\left(\frac{1}{2N}\right)$

Simulate $t_3 \sim \text{Exp}\left(\frac{3}{2N}\right)$

Simulate $t_4 \sim \text{Exp}\left(\frac{3}{N}\right)$

Simulate $t_5 \sim \text{Exp}\left(\frac{5}{N}\right)$

This concept is easy to extend to multiple populations with splits and migrations.

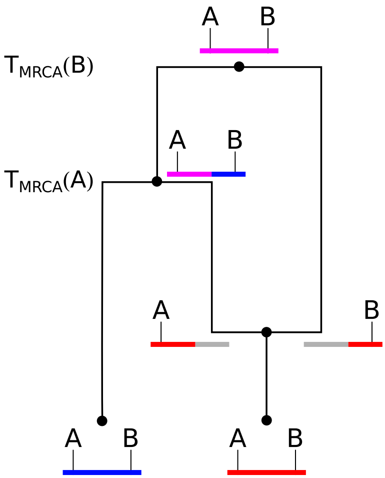
Ancestral recombination graphs (ARG): a genealogical interpretation of LD

Recombination splits lineages

Going backward in time, a recombination event splits a lineage into two independent lineage, each with a part of the locus.

Coalescenes merges lineages

In contrast, coalescent events merge lineages together.



Ancestral recombination graphs (ARG): a genealogical interpretation of LD

Probability that two loci have the same genealogy

Per generation, we have:

$\mathbb{P}(\text{coalescent event}) = \frac{1}{2N}$

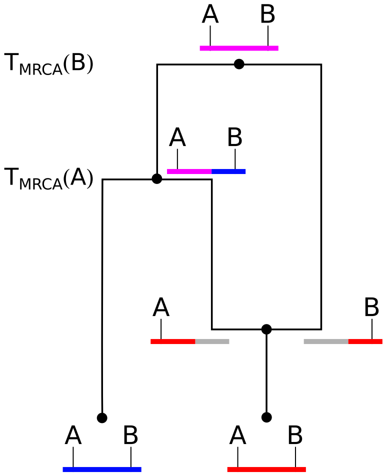
$\mathbb{P}(\text{recombination event}) = 2c.$

$\mathbb{P}$(coalescencne before recombination)

$$= \frac{\frac{1}{2N}}{\frac{1}{2N} + 2c} = \frac{1}{1 + 4Nc}$$

Example

If $N = 10^6$ and $c = 0.01$ per Mb, the probability that two loci that are 1Kb apart have the same tree is thus $\frac{1}{1+4\cdot10^6\cdot10^{-5}} = \frac{1}{1+4\cdot10} \approx 2.4\%$



Summary

- Coalescent theory models the evolutionary history backward in time. Allows to characterize the diversity of a sample efficiently.
- The expected number time to the MRCA of two samples is $2N$ generations. The expected number of mutations is $\theta = 4N\mu$. But there is considerable variation.
- Among a large sample, the expected time to the MRCA is $4N$. Older times are observed in case of population substructure.
- Coalescent theory also allows us to simulate genetic diversity efficiently.